



LECTURE

Introduction to **HashiCorp Vault**

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Learning Objectives

By the end of this module, you will:



Understand what HashiCorp Vault is and the benefits of using it



Be able to understand the problems that Vault aims to solve



Know the differences between available versions of Vault



How to install Vault on a host machine

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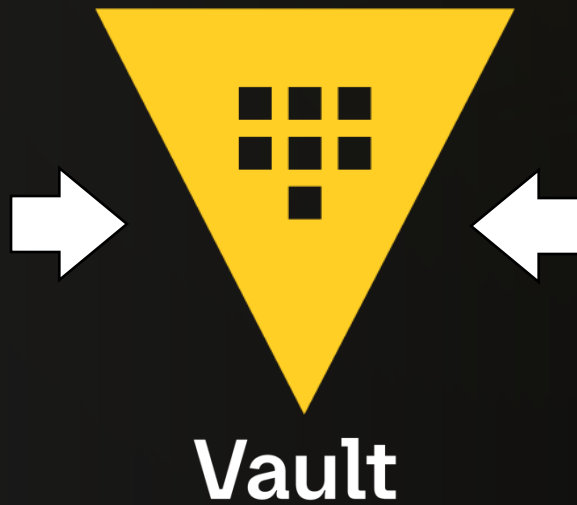


What is HashiCorp Vault?

- Vault is a tool for securing and accessing secrets
- It helps manage sensitive data such as passwords, API keys, certifications, and more
- Provides secure storage, access control, and audit logging

Consumption

End users
Engineers
Pipelines
Applications



Sensitive Data

Passwords
Certificates
API keys
Service accounts



What is HashiCorp Vault?

Beyond passwords, Vault is a fully-functional security platform that provides additional functionality such as:

- Centralized encryption and tokenization services
- Automated generation of PKI certifications
- Dynamic credentials for accessing cloud platforms and databases
- Secure injection of secrets for Kubernetes
- Standardized workflow for key management across KMS providers



Benefits of HashiCorp Vault



Store Long-Lived Static
Secrets



Dynamically Generate
Secrets, upon Request



Fully-Featured API



Identity-based Access



Provide Encryption
as a Service



Act as a Root or
Intermediate Certificate
Authority



What Problems Does Vault Solve?

- Eliminates hardcoded secrets in code
- Eliminates secret sprawl
- Ensures secrets are only accessible by authorized entities (governance)
- Manages the lifecycle of secrets by dynamically generating them only when they are needed
- Enables detailed audit logging for compliance.



Comparing Versions of **HashiCorp Vault**



HashiCorp Vault – Compare Versions

- **Community Version:** Free, community-supported version
- **Vault Enterprise:** Paid version with additional features and support.
- **HashiCorp Cloud Platform (HCP):** Hosted version of Vault (SaaS)
- **Vault Secrets:** Cloud-based service for storing credentials
- **Vault Radar:** Cloud-based service to scan and discover secrets

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HashiCorp Vault – Compare Versions

SELF-HOSTED

FREE

COMMUNITY

Static & Dynamic Secrets

Encryption Key Rotation

Detailed Audit Logs

Vault Agent

High Availability (Clustering)

Access Control Policies

Encryption as a Service

AWS, Azure, & GCP Auto-Unseal

VS

ENTERPRISE

Resource Quotas

Namespaces

Replication

Performance Standby Nodes

HSM Auto-Unseal

Additional Secrets Engines

Secrets Sync

FIPS 140-2 & Seal Wrap

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HashiCorp Vault – Compare Versions

CLOUD-BASED SERVICES

Vault Dedicated (HCP)

- Hosted by HashiCorp
- Fully Managed Solution
- Reduce Admin Burden
- Scalable
- Push Button Deployment
- Pay by the Hour
- All Enterprise Features
- Dev or Prod Options

Vault Secrets

- Hosted by HashiCorp
- Centralized Secrets
- Access from Anywhere
- Sync with Cloud-Based Services

Vault Rader

- Hosted by HashiCorp
- Scan for Stored Secrets
- NOT a Secrets Manager
- Acquired via BluBracket

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**This course will focus on the
functionality and services provided
by self-hosted Vault.**



Interacting with **Vault**

Interacting with Vault

Vault Interfaces

- Vault provides a fully-functionality HTTP API, a CLI, and a UI
- Not all Vault features are available via UI and CLI, but all features can be accessed using the HTTP API
- The Vault UI must be enabled in the configuration file

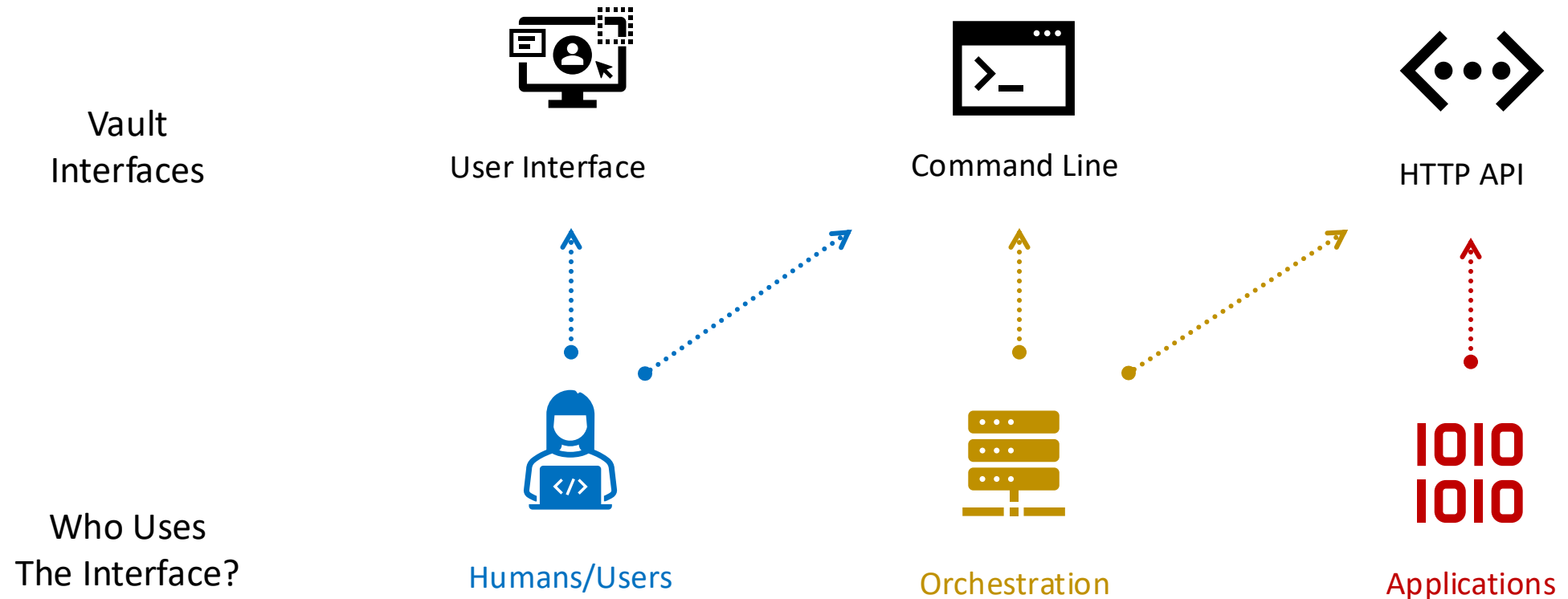
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Vault Interfaces

Who Uses the Vault Interface?



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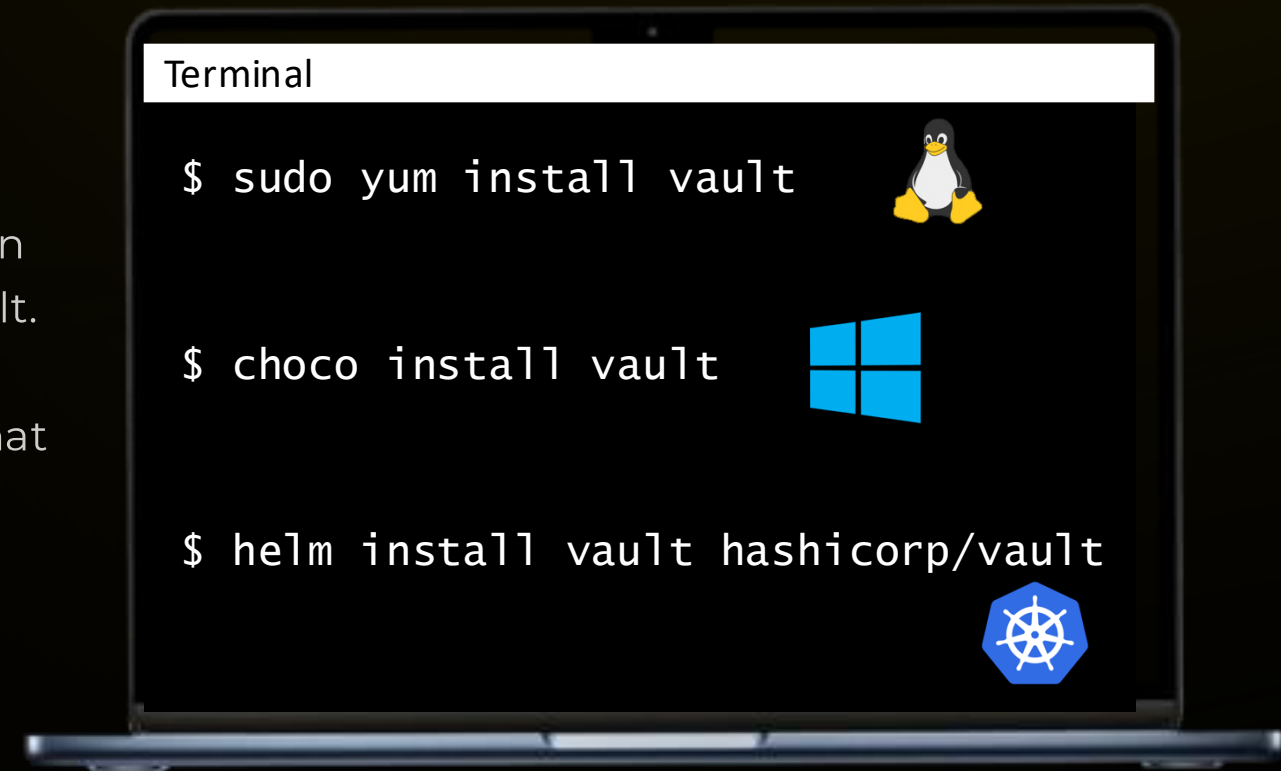


Installing HashiCorp Vault



Installing HashiCorp Vault

- Vault is a single binary that provides all functionality for each version
- Since Vault is platform agnostic, installation can vary depending on where you want to run Vault.
- Most platforms support a package manager that can be used to install the Vault binary and prerequisites for running Vault as a service.
- Helm can be used to simplify Kubernetes implementations





Installing HashiCorp Vault

- Once installed, a configuration file is needed to setup critical components of Vault, such as storage, encryption, etc.
- When Vault is started, Vault uses this file to determine how to access data, what interface to listen on, how to unseal, and more.

```
1 storage "raft" {
2   path      = "/opt/vault/data"
3   node_id   = "node-a-us-east-1.example.com"
4   retry_join {
5     auto_join = "provider=aws region=us-east-1 tag_key=vault tag_value=us-east-1"
6   }
7 }
8 listener "tcp" {
9   address = "0.0.0.0:8200"
10  cluster_address = "0.0.0.0:8201"
11  tls_disable = 0
12  tls_cert_file = "/etc/vault.d/client.pem"
13  tls_key_file = "/etc/vault.d/cert.key"
14  tls_disable_client_certs = "true"
15 }
16 seal "awskms" {
17   region = "us-east-1"
18   kms_key_id = "12345678-abcd-1234-abcd-123456789101",
19   endpoint = "example.kms.us-east-1.vpce.amazonaws.com"
20 }
21
22 api_addr = "https://vault-us-east-1.example.com:8200"
23 cluster_addr = "https://node-a-us-east-1.example.com:8201"
24 cluster_name = "vault-prod-us-east-1"
25 ui = true
26 log_level = "INFO"
27 license_path = "/opt/vault/vault.hcl"
28 disable_mlock=true
```

github.com/btkrausen/hashicorp

DEMO

Installing

HashiCorp Vault

